IMPERIAL

# A Musical Application of Cluster Analysis

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## Introduction

Cluster analysis is a technique in which observations are partitioned into clusters.

There is a stigma around Taylor Swift's music, which causes her music to be perceived as plain. In this project, cluster analysis is used to show that there exists variety within her style.

## Dataset

The data used is a subset of Swift's works, obtained from Kaggle (Priester, 2024). [1]

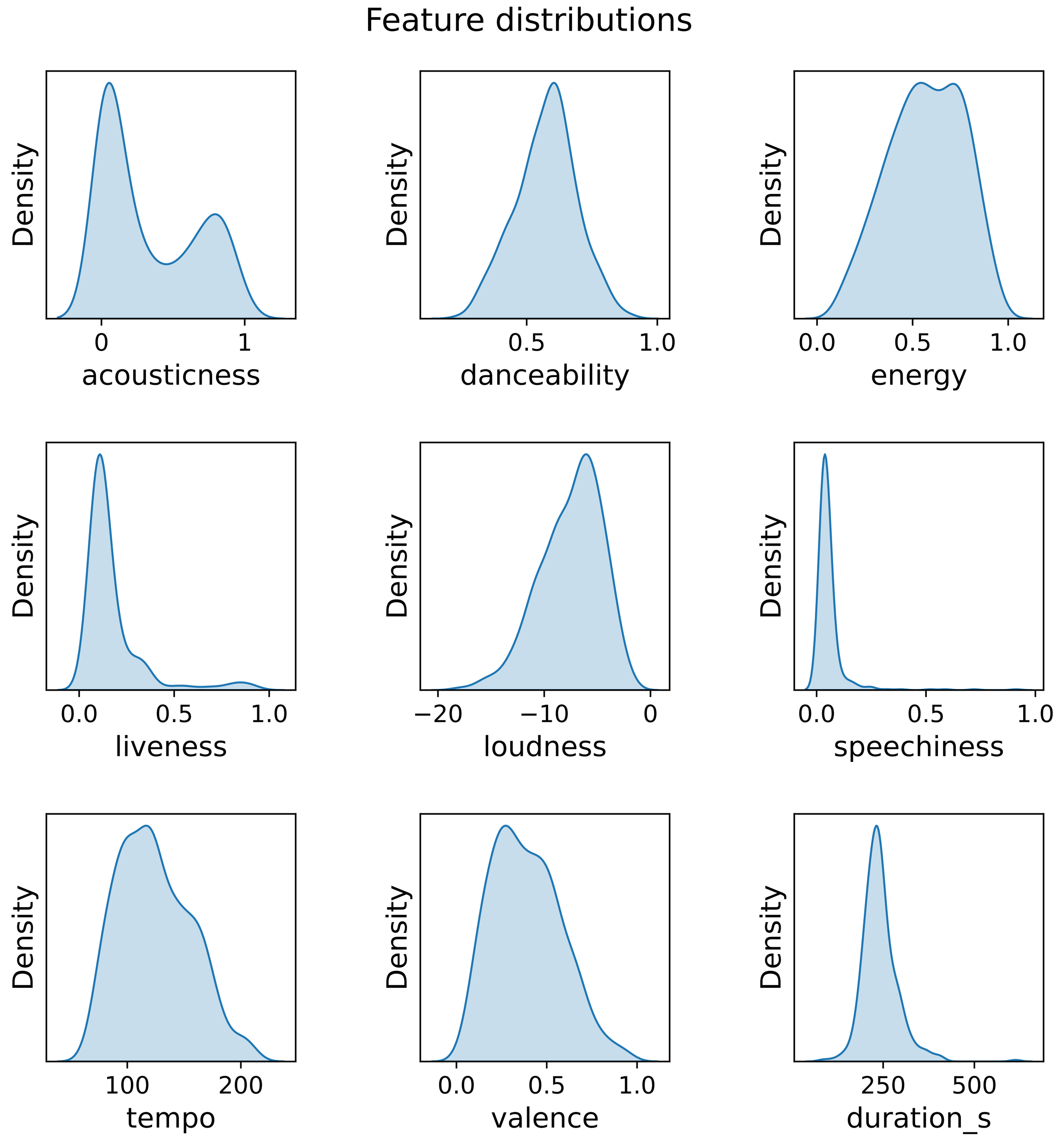


**Figure 1:** Density estimate plots of musical features.

## Code

Clustering and visualisations were done in Python, using numpy, pandas, scikit-learn, matplotlib and seaborn. The code used can be found here: https://github.com/lewistang-uk/Cluster-Analysis/

## Method

- Scale features to standard normal distribution.
- Select number of clusters using intrinsic metrics.
- Apply clustering algorithm using all features.
- Select features by analysing distributions across clusters.
- Apply clustering algorithm using selected features.
- Rename clusters by their main characteristic.

## Selection of number of clusters

- Elbow (Thorndike, 1953) [2]
  - Plot WCSS (Within-Cluster Sum of Squares) against number of clusters.
  - Find the point where the gradient drops sharply.
- Silhouette (Rousseeuw, 1987) [3]
  - Measure the similarity of a point to its own cluster compared to other clusters.
  - Find the average of this metric and maximise by varying the number of clusters.

## Clustering algorithms

- K-Means Clustering (MacQueen, 1967) [4]
  - Initialise by selecting cluster centres (centroids).
  - Assign each observation to a cluster by its nearest centroid.
  - Adjust centroid position to minimise within-cluster variance.
  - Iterate until convergence.
- Agglomerative Clustering (Ward, 1963) [5]
  - Initialise with each observation in its own cluster.
  - Recursively merge the most similar pair of clusters until the required number of clusters is obtained.

## Application of clustering

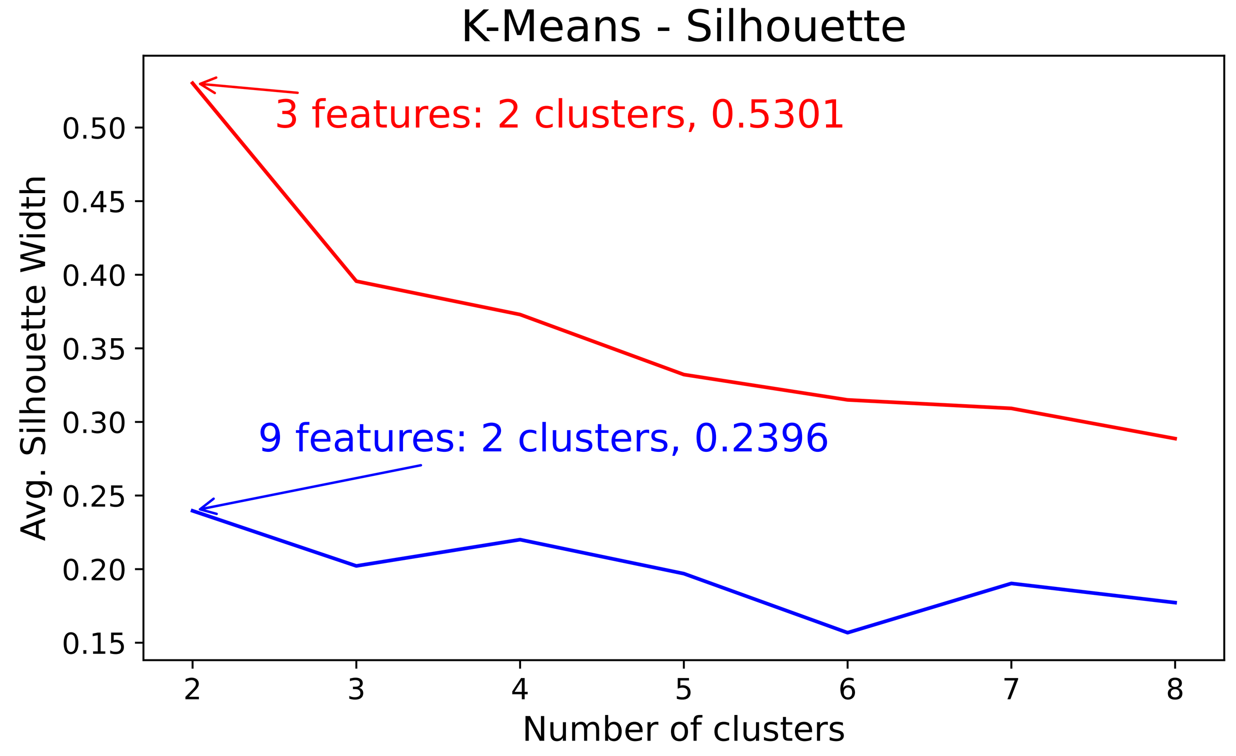


**Figure 2:** Feature selection reduces noise, creating clearer clusters. This significantly increases average silhouette width while suggesting the same number of clusters.

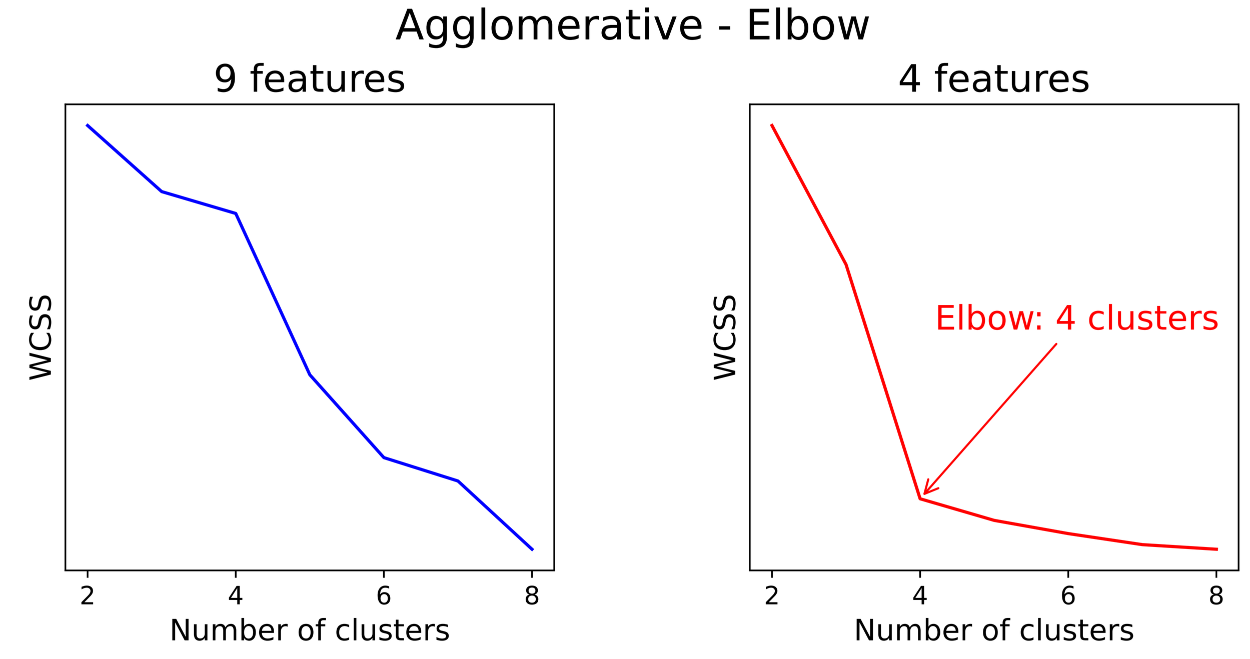


**Figure 3:** After feature selection, an elbow is present in the WCSS graph.

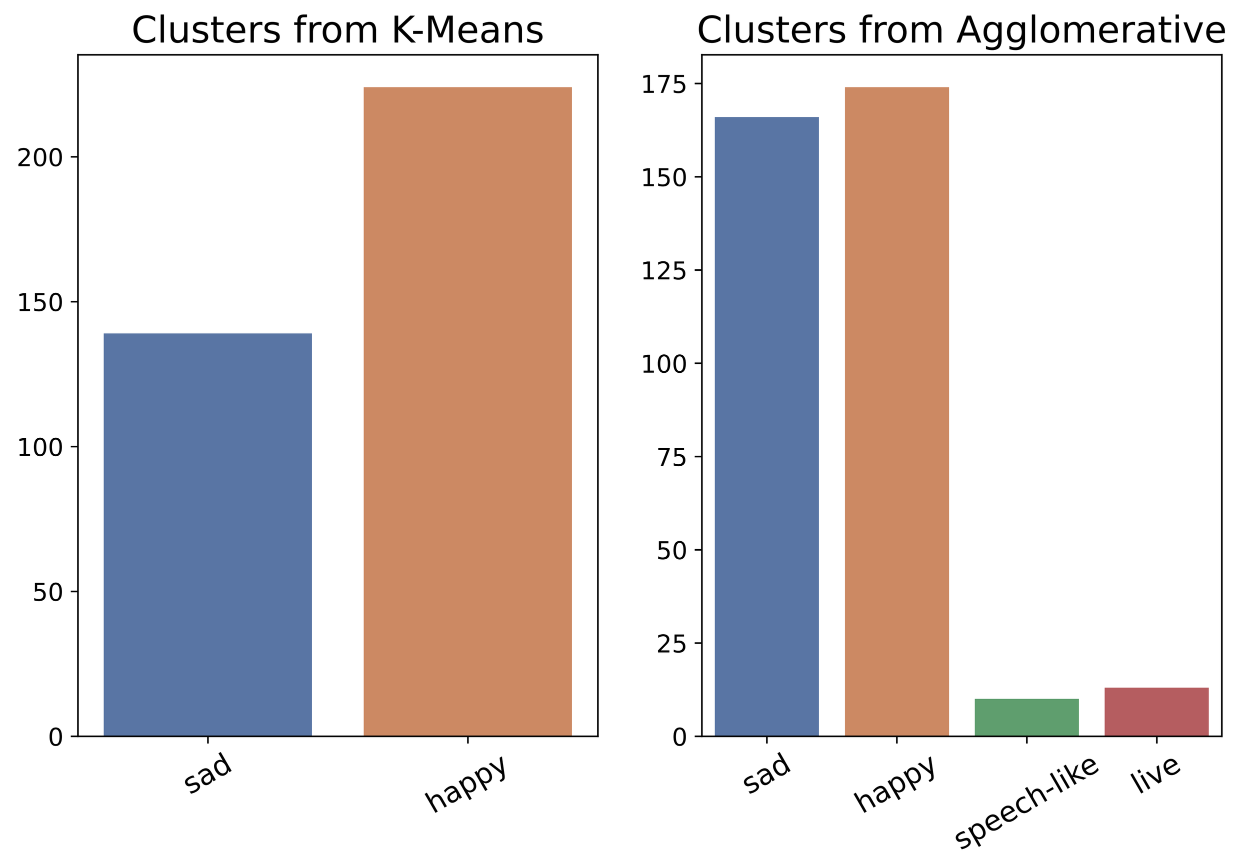


**Figure 4:** Number of observations in each cluster for both clustering algorithms.

## Examples of differing cluster labels

| song | K-Means | Agglomerative |
|---|---|---|
| closure | sad | speech-like |
| Treacherous | happy | sad |
| BTR - Live/2011 | happy | live |

## Summary

- Agglomerative finds more hidden patterns – K-Means implicitly assumes that clusters are roughly equal in cardinality (Bishop, 2006). [6]
- Silhouette is a better evaluation metric than Elbow – the implied optimum can always be found.
- Both clustering algorithms imply two main clusters, showing that there is some variety within Swift's music.
- In an interview, Swift (2022) [7] described three different styles of lyric writing. Analysis of lyrics could point to a different clustering.

## References

[1] Priester, J (2024), Taylor Swift Spotify Dataset, Version 21. Retrieved 24/05/26 from www.kaggle.com/datasets/jarredpriester/taylor-swift-spotify-dataset/ [2] Thorndike, R.L. (1953) Who belongs in the family?. *Psychometrika*. 18(4), 267–276. [3] Rousseeuw, P.J. (1987) Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*. 20(1), 53–65. [4] MacQueen, J. (1967) Some methods for classification and analysis of multivariate observations. *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability*. 1, 281–297. [5] Ward, J.H. (1963) Hierarchical grouping to optimize an objective function. *Journal of the American Statistical Association*. 58 (301), 236–244. [6] Bishop, C.M. (2006) *Pattern Recognition and Machine Learning*. New York, Springer. [7] Rogerson, B. (2022) Taylor Swift reveals the 3 lyric 'genre categories' she uses when she's writing songs. MusicRadar, 21 September. Retrieved 08/06/26 from www.musicradar.com/news/taylor-swift-songwriting-secret